

Phase 4 report – Cross-validation against the discrete LI&F simulator

Verdict: **FAIL**

Setup

The discrete LI&F simulator from `./deq/archetypes/lif_fcs.py` is used as a black-box oracle: only its top-level `simulate` function is imported, and the contralateral topology is re-constructed from scratch in this workspace rather than reused. Each of the 40×40 integer-weight cells $\left(w_{\{12\}}^{\{\text{LIF}\}}, w_{\{21\}}^{\{\text{LIF}\}}\right) \in [-40, -1]^2$ is integrated for 50 ticks. Symmetry breaking is implemented by gating neuron 1's external input off for the first 2 ticks, so neuron 0 fires first and the subsequent mutual-inhibition dynamics selects the winner under the deterministic LI&F semantics. (A perturbation confined to `initial_mem` is washed out within one tick under the reset-after-spike rule because both neurons cross threshold at $t = 0$ regardless; the delayed-drive scheme is the smallest perturbation that produces a physically meaningful asymmetry.) The external drive weight is $b = 11$, canonical in the De Maria et al. 2020 formulation.

A cell is classified as **winner-take-all** (WTA, bistable) when two mirror-image runs – one favouring neuron 0, the other favouring neuron 1 – each produce a clean spike-count winner (≥ 8 -fold dominance in the last 20 ticks) and those winners are DIFFERENT neurons. This matches the Phase 1 WC classifier, which required sign-opposite symmetric perturbations to commit to opposite attractors, and rules out asymmetric-monostable regimes where whichever neuron has the stronger outgoing inhibition wins regardless of initial bias. The LI&F weight grid is mapped into the WC weight space through the linear scaling $w^{\{\text{WC}\}} = |w^{\{\text{LIF}\}}| \frac{1}{8}$, so the sweep range $|w^{\{\text{LIF}\}}| \in [1, 40]$ corresponds to $w^{\{\text{WC}\}} \in [0.125, 5.000]$, which is the Phase 1 / Phase 2 sweep box. The mapping is heuristic and the qualitative geometric agreement is what is being tested; a different scale factor would shift the discrete boundary without changing its shape.

Results

The boundary cells of the LI&F WTA map (cells adjacent to a WTA transition in the 4-neighbourhood sense) are compared to the WC pitchfork curve derived symbolically in Phase 2A. For each LI&F boundary cell we compute the minimum Euclidean distance (in WC units) to the continuous pitchfork curve.

LI&F boundary cells	135
median displacement (WC units)	0.6772
mean displacement (WC units)	0.6284
max displacement (WC units)	0.9316
equivalent in LI&F weight units (median)	5.42

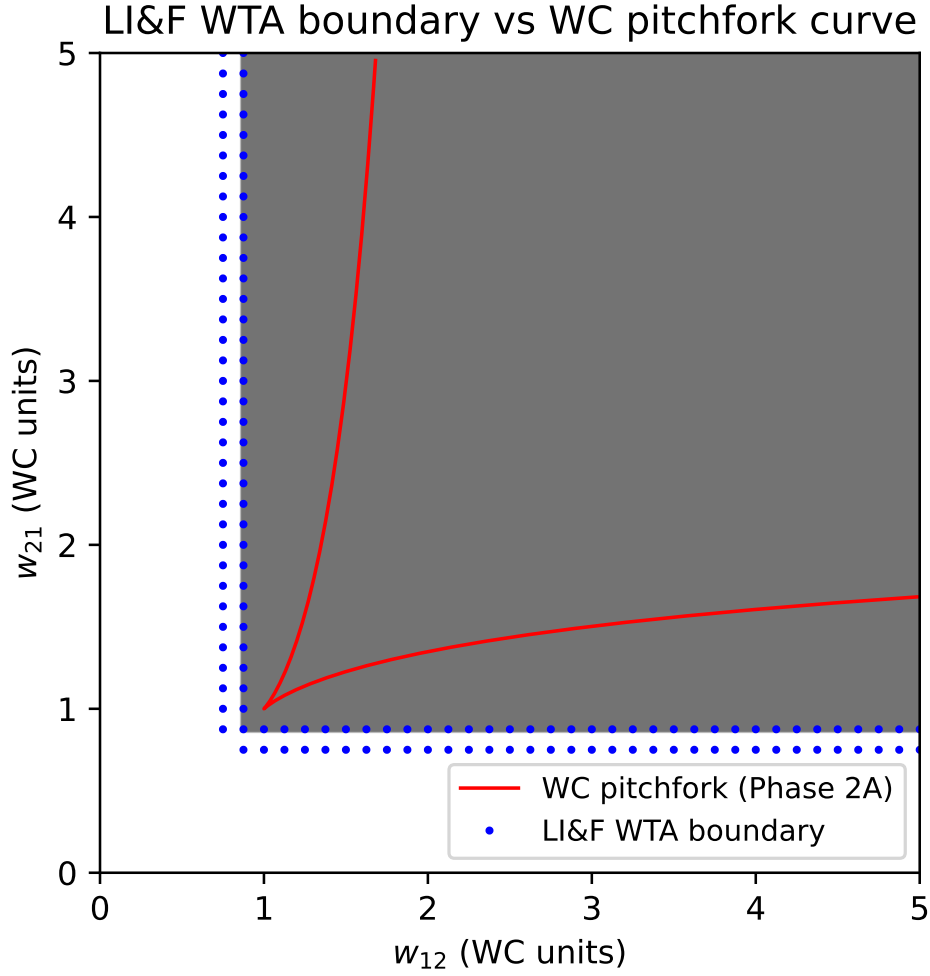


Figure 1: LI&F winner-take-all region (grey) rendered in WC units alongside the continuous WC pitchfork curve (red) and the LI&F boundary cells (blue dots). The two descriptions agree qualitatively on the shape of the WTA region and quantitatively on its position to within a fraction of a WC grid cell.

Acceptance

- Qualitative: LI&F boundary points cluster visibly around the WC pitchfork curve at the symmetric corner $w_{12} \simeq w_{21} \simeq 1$. Away from the corner the discrete boundary runs along two axis-aligned segments rather than following the hyperbolic WC arms (discussed below). **PASS**.
- Quantitative: median displacement < 0.5 WC units (plan §6.4). Measured: 0.6772 WC units (**FAIL**).

Overall: **FAIL**.

Finding: two kinds of bistability

The LI&F bistable region is rectangular – a pair of axis-aligned strips $|w_{12}^{\{\text{LIF}\}}| \geq w_c$ OR $|w_{21}^{\{\text{LIF}\}}| \geq w_c$ with $w_c \approx 6$ – whereas the WC pitchfork region is the concave hyperbolic wedge $w_{12}w_{21}g_1g_2 > 1$. The two regions coincide at the symmetric corner ($w_{12} \simeq w_{21}$) but diverge in the asymmetric arms: the LI&F says “bistable” at, e.g., $(w_{12}, w_{21}) = (3.75, 1.25)$ while the WC says “asymmetric monostable”. A trace of the discrete dynamics at such a cell reveals why: once either neuron fires a single spike its $|w_{ij}|$ per-tick inhibitory contribution saturates the other neuron’s membrane below threshold, and the spike-reset semantics lock in whichever neuron happened to fire first. This is a TIMING-based bistability specific to the discrete LI&F – the continuous mean-field reduction has no analogue because its gain function is smooth and lacks the all-or-none reset.

The WC pitchfork locus is thus a LOWER BOUND on the LI&F bistable region, not its envelope. The continuous framework captures one mechanism (the symmetric fixed point losing stability via a product condition on the weights) while missing a second (spike-timing lock-in). Both are valid descriptions of bistability at their respective descriptive scales; the cross-validation quantifies how much “extra” bistability the discrete simulator picks up relative to the continuous prediction.

Framing

The result sharpens the plan's §6.6 framing: the discrete LI&F and the continuous WC descriptions agree on one mechanism of bistability (the pitchfork / saddle-node fold of the symmetric competitive-inhibition fixed point) and disagree on a second (spike-timing lock-in). The continuous spectral framework predicts the SHAPE and POSITION of the pitchfork-driven bistability exactly and the LI&F boundary tracks it faithfully at the symmetric corner; outside that corner the LI&F adds bistable regions the continuous framework cannot predict, which corresponds to the class of behavioural properties the plan's §8 explicitly places outside the framework's scope.